

Conditional Expectation, Introductory Inference

Before you leave lab, make sure you [click here](#) so that you're marked as having attended this week's section. The CA leading your discussion section can enter the password needed once you've submitted.

Warm-ups

1 Why Multiple Random Variables?

What is a probabilistic model with multiple random variables? What does the term inference mean? What do you call the probability of an assignment to all variables in a probabilistic model? Why is that useful? Why can it be hard to represent?

A probabilistic model is a way of defining the relationship between many random variables. Inference is the act of computing a belief in one (or more) variables based on an observation. The probability of an assignment to all variables in a probabilistic model is called the joint. The joint can be used to solve any inference task. The number of ways of assigning values to variables is exponential in the number of random variables.

2 Joint Random Variables Statistics

True or False? The symbol Cov is covariance, the symbol $\wedge$ is logical-and, the symbol ρ is Pearson correlation, the symbol $\implies$ is logical implication, and $X \perp Y$ is just a fancy way to say that X and Y are independent.

A statement like " $A \sim \text{Bin}(10, 0.5) \wedge B \sim \text{Bin}(10, 0.5) \wedge A \perp B \implies A + B \sim \text{Bin}(20, 0.5)$ " reads "If A and B are both distributed as Binomials with the same parameters, then $A + B$ is a Binomial as well with the same p parameter and an n parameter that's the sum of the those for A and B ."

$X \perp Y \implies Cov(X, Y) = 0$	$Var(X + X) = 2Var(X)$
$Cov(X, Y) = 0 \implies X \perp Y$	$X \sim \mathcal{N}(0, 1) \wedge Y \sim \mathcal{N}(0, 1) \implies \rho(X, Y) = 1$
$Y = X^2 \implies \rho(X, Y) = 1$	$Y = 3X \implies \rho(X, Y) = 3$

True or False?

True	False (... = $4Var(X)$)
False (antecedent necessary, not sufficient)	False (don't know how independent X & Y are)
False ($Y = X \implies \dots$)	False (... = 1)

3 Random Number of Random Variables

Let N be a non-negative integer-valued random variable—that is, a random variable that takes on values in $\{0, 1, 2, \dots\}$. Let $X_1, X_2, X_3, \dots$ be an infinite sequence of independent and identically distributed random variables (independent of N), each with mean μ , and $X = \sum_{i=1}^N X_i$ be the sum of the first N of them.

Before doing any work, what do you *think* $E[X]$ will turn out to be? Then show it mathematically to see if your intuition is correct.

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^N X_i\right] = \sum_n E\left[\sum_{i=1}^N X_i \mid N = n\right] p_N(n) = \sum_n E\left[\sum_{i=1}^n X_i \mid N = n\right] p_N(n) \\ &= \sum_n E\left[\sum_{i=1}^n X_i\right] p_N(n) = \sum_n n\mu p_N(n) = \mu \sum_n n p_N(n) = \mu E[N] \end{aligned}$$

Alternatively,

$$E[X] = E[E[X|N]] = E[N\mu] = \mu E[N]$$

Problems

1 CS106A Is Popular

CS106A is Stanford's introductory programming course and largely considered the primary gateway to our undergraduate major. Assume next quarter's offering of CS106A is exactly 600 people, that each of the four undergraduate classes is compromised of 1750 students, and that next quarter's CS106A roster is just some random sample of the 7000 undergraduates. Let A, B, C , and D count the number of freshman, sophomores, juniors, and seniors in the class of 600.

- Present the joint probability mass function of A, B, C, D ? Restated, present an expression for $P(A = a, B = b, C = c, D = d)$.

$$P(A = a, B = b, C = c, D = d) = \frac{\binom{1750}{a} \binom{1750}{b} \binom{1750}{c} \binom{1750}{d}}{\binom{7000}{600}}$$

- Does your PMF from part a) describe a multinomial random variable? Intuitively justify your answer.

No, a multinomial random variable requires that trials be independent. If we define a trial in this case as picking one student to be in CS106A, independence would require us to select students with replacement, which isn't the case here.

- What is the conditional probability distribution of A given that $B + C = 300$? Restated, what is $P(A = a | B + C = 300)$?

This is the same type of problem as that in part a), but we are now distributing 300 students across two groups, freshmen and seniors.

$$P(A = a | B + C = 300) = \frac{\binom{1750}{a} \binom{1750}{300-a}}{\binom{3500}{300}}$$

- Do you expect $Cov(A, B)$ to be positive, zero, or negative? Justify your answer.

We expect $Cov(A, B)$ to be negative. If there are more freshmen in the class, there will be fewer spots left for sophomores. Incidentally, the $\rho(A, B)$ value—that is, the normalized covariance—would be very, very slightly negative because of the large number of students and the large class size.

2 Managing Screen Time

Push notifications light up our phones at a rate that's guided by a Poisson process with an constant average rate of 5 notifications per hour at all hours, night and day. In an effort to maximize productivity, you put your phone down and ignore it as much as possible. You do, however, periodically check it to clear all notifications. You check at 7am when you wake up, noon when you grab lunch, 5pm as you wrap up classes for the day, and then again at 10pm before you go to sleep.

Let $W, X, Y,$ and Z be Poisson random variables that count the number of push notifications that have accumulated from 10pm to 7am, 7am to noon, noon to 5pm, and 5pm to 10pm, respectively. We assume that the number of push notifications that arrive within each interval are all mutually independent of all other intervals.

- Compute the joint PMF on $W, X, Y,$ and Z .

Since all four variables are independent, the joint PMF is the product of the four single-dimensional PMFs. That means that:

$$P(W = w, X = x, Y = y, Z = z) = \frac{e^{-45} 45^w}{w!} \frac{e^{-25} 25^x}{x!} \frac{e^{-25} 25^y}{y!} \frac{e^{-25} 25^z}{z!}$$

Ship it!

- Compute the conditional joint PMF on $W, X, Y,$ and Z given that $W + X + Y + Z = 150$.

When adding Poisson random variables, it's nice to have another random variable modelling that sum. Let's define $S = W + X + Y + Z$ with the understanding that the sum of independent Poisson random variables is itself a Poisson where the one parameter is the sum of the individual ones. That means $S \sim \text{Poi}(120)$ and that $P(S = s) = \frac{e^{-120} 120^s}{s!} = \frac{e^{-120} 120^{w+x+y+z}}{s!}$.

What's the conditional joint PMF of interest? Check this out:

$$\begin{aligned}
 P(W = w, X = x, Y = y, Z = z | S = s) &= \frac{P(S = s, W = w, X = x, Y = y, Z = z)}{P(S = s)} \\
 &= \frac{P(W = w, X = x, Y = y, Z = z)}{P(S = s)} \\
 &= \frac{\frac{e^{-45} 45^w}{w!} \frac{e^{-25} 25^x}{x!} \frac{e^{-25} 25^y}{y!} \frac{e^{-25} 25^z}{z!}}{\frac{e^{-120} 120^s}{s!}} \\
 &= \frac{\frac{45^w}{w!} \frac{25^x}{x!} \frac{25^y}{y!} \frac{25^z}{z!}}{\frac{120^{w+x+y+z}}{s!}} \\
 &= \binom{s}{w, x, y, z} \left(\frac{9}{24}\right)^w \left(\frac{5}{24}\right)^x \left(\frac{5}{24}\right)^y \left(\frac{5}{24}\right)^z
 \end{aligned}$$

The transition from the first line to the second is just stating that $S = s$'s presence in the joint PMF is redundant, since $s = w + x + y + z$. The bottom line here is that the conditional PMF devolves into a multinomial random variable. When s is 150, our probability is just

$$P(W = w, X = x, Y = y, Z = z | S = 150) = \binom{150}{w, x, y, z} \left(\frac{9}{24}\right)^w \left(\frac{5}{24}\right)^x \left(\frac{5}{24}\right)^y \left(\frac{5}{24}\right)^z$$

- Compute the conditional PMF of $X + Y + Z$ —that's the number of notifications that arrive while you're awake—given that $W + X + Y + Z = 150$.

This can be derived from first principles much as we derived the conditional joint PMF in the previous part to this question. Or we can simply merge the three intervals of the multinomial to arrive at a binomial, i.e., $X + Y + Z | S = 150 \sim \text{Bin}(150, \frac{15}{24}) = \text{Bin}(150, \frac{5}{8})$.

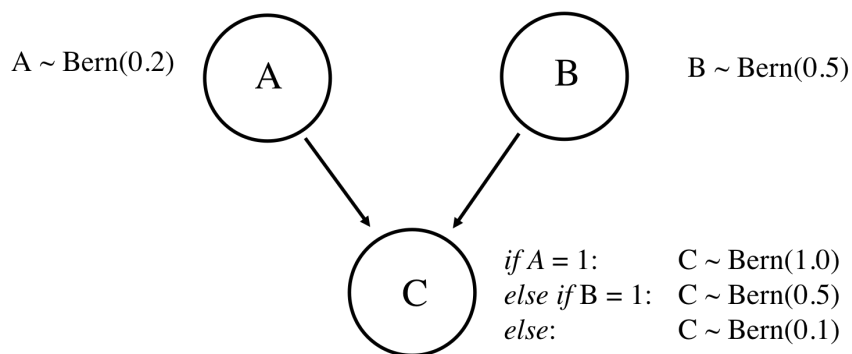
- Compute $E[X + Y + Z | W + X + Y + Z = 150]$ and $\text{Var}(X + Y + Z | W + X + Y + Z = 150)$.

$$E[X + Y + Z | W + X + Y + Z = 150] = 150 \cdot \frac{5}{8} = 93.75 \text{ and } \text{Var}(X + Y + Z | W + X + Y + Z = 150) = 150 \cdot \frac{5}{8} \cdot \frac{3}{8} = 35.15625$$

3 Understanding Bayes Nets

	A = 0		A = 1	
	B = 0	B = 1	B = 0	B = 1
C = 0	0.36	0.20	0.00	0.00
C = 1	0.04	0.20	0.10	0.10

The **joint probability table (above)** for random variables A , B and C is equivalent to the **Bayes network (below)**. Both give the probability of any combination of the random variables. In the Bayes network the probability of each random variable is provided given its causal parents.



- Use the Bayes network to explain why $P(A = 0, B = 1, C = 1) = 0.20$

$$P(A = 0, B = 1, C = 1) = P(A = 0)P(B = 1)P(C = 1|A = 0, B = 1) = 0.8 * 0.5 * 0.5 = 0.2.$$

- What is $P(A = 1|C = 1)$?

Using the table, we see that

$$P(A = 1|C = 1) = \frac{0.1 + 0.1}{0.1 + 0.1 + 0.2 + 0.04} = \frac{0.2}{0.44} = \frac{5}{11}$$

- Is A independent of B ? Explain your answer.

Yes. This follows directly from the structure of the bayesian network, because A and B have no shared ancestors. Alternatively, note that $P(A = a, B = b) = P(A = a)P(B = b)$, which satisfies the definition of independence.

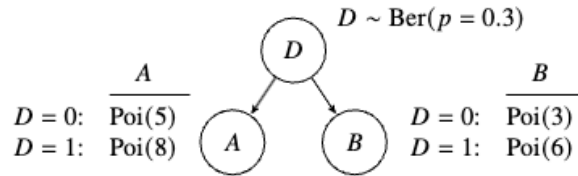
- Is A independent of B **given** $C = 1$? Explain your answer.

No. From the table, we can see that $P(B = 1|A = 0, C = 1) = \frac{0.2}{0.4+0.2} \neq P(B = 1|A = 1, C = 1) = \frac{0.1}{0.1+0.1}$. So given $C = 1$, knowing the value of A informs us about the value of B , and therefore A and B are not conditionally independent given C .

Note: This phenomenon is sometimes called "Explaining Away" if you're curious to read more.

4 Fish Sticks [courtesy of Lisa Yan]

Fish Sticks, the online platform designed to meet all of your fish stick needs, wants to model their hourly homepage traffic from Stanford. The company decides to model two different behaviors for homepage visits according to the Bayesian Network below:



A and B are the numbers of Stanford students and faculty, respectively, who visit the Fish Sticks homepage in an hour. Since Fish Sticks does not know when Stanford people eat, the company models demand as a "hidden" Bernoulli random variable D , which determines the distribution of A and B . Recall that in a Bayesian Network, random variables are conditionally independent given their parents. For example, given $D = 0$, $A \sim \text{Poi}(5)$ and $B \sim \text{Poi}(3)$, two independent random variables.

- a. Given that 6 users from group A visit the homepage in the next hour, what is the probability that $D = 0$?

Note that given $D = 0$, $A \sim \text{Poi}(\lambda = 5)$, and given $D = 1$, $A \sim \text{Poi}(\lambda = 8)$. By Bayes' Theorem,

$$\begin{aligned}
 P(D = 0|A = 6) &= \frac{P(A = 6|D = 0)P(D = 0)}{P(A = 6|D = 0)P(D = 0) + P(A = 6|D = 1)P(D = 1)} \\
 &= \frac{\frac{5^6 e^{-5}}{6!}(1 - 0.3)}{\frac{5^6 e^{-5}}{6!}(1 - 0.3) + \frac{8^6 e^{-8}}{6!}(0.3)} \\
 &= \frac{5^6 e^{-5}(1 - 0.3)}{5^6 e^{-5}(1 - 0.3) + 8^6 e^{-8}(0.3)} \approx 0.7364
 \end{aligned}$$

- b. What is the probability that in the next hour, the *total* number of users who visit the homepage from groups A and B is equal to 12, i.e., what is $P(A + B = 12)$?

By Law of Total Probability,

$$P(A + B = 12) = P(A + B = 12|D = 0)P(D = 0) + P(A + B = 12|D = 1)P(D = 1).$$

A and B are conditionally independent Poisson random variables given D , and therefore $A + B|D = 0 \sim \text{Poi}(\lambda = 8)$ and $A + B|D = 1 \sim \text{Poi}(\lambda = 14)$. Using the Poisson PMF,

$$P(A + B = 12) = \frac{8^{12}e^{-8}}{12!} \cdot (1 - 0.3) + \frac{14^{12}e^{-14}}{12!} \cdot (0.3) \approx 0.0632.$$

c. Now simulate $P(A + B = \text{total})$, where $\text{total} = 12$, by implementing the `infer_prob_total(total, ntrials)` function below using rejection sampling.

- `total` is the total number of users from groups A and B in the event $A + B = \text{total}$.
- `ntrials` is the number of observations to generate for rejection sampling.
- `prob` is the return value to the function, where $\text{prob} \approx P(A + B = \text{total})$.
- The function call is implemented for you at the bottom of the code block.

You can call the following functions from the `scipy` package:

- `stats.bernoulli.rvs(p)`, which randomly generates a 1 with probability p , and generates a 0 otherwise.
- `stats.poisson.rvs( $\lambda$ )`, which randomly generates a value according to a Poisson distribution with parameter λ

You are not required to use lists or `numpy` arrays in this question (but you can if you want).

Pseudo-code is fine as long as your code accurately conveys your approach.

```
import numpy as np
from scipy import stats

def infer_prob_total(total, ntrials):
    # here's where your implementation belongs
    return prob

total = 12
ntrials = 50000
print('Simulated% P(A + B)=', infer_prob_total(total, ntrials))
```

This is the full implementation right here:

```
import numpy as np
from scipy import stats

def infer_prob_total(total, ntrials):
```

```

n_samples_event = 0
for i in range(ntrials):
    d = stats.bernoulli.rvs(0.3)
    if d == 0:
        user_sum = stats.poisson.rvs(5) + stats.poisson.rvs(3)
    else:
        user_sum = stats.poisson.rvs(8) + stats.poisson.rvs(6)
    if user_sum == total: n_samples_event += 1
prob = n_samples_event/ntrials
return prob

ntrials = 50000
total = 12
print("Simulated P(A + B)=", infer_prob_total(total, ntrials))

```

5 ChatGPT, Watermarking, and Bayesian Inference

ChatGPT is a generative AI technology that can be coarsely summarized to be a chat bot with a seemingly boundless ability to discuss any topic—history, computer science, art, nuclear physics, probability, and even the ethics of using ChatGPT—in any one of several written languages, including English, French, Spanish, C++, JavaScript, Python, and some 100 others.

Unsurprisingly, we will soon be left to wonder whether a poem, a Tweet, a C++ function, or a college thesis is written by ChatGPT or a human being. Questions about authorship, accuracy, and attribution have prompted OpenAI, the company behind ChatGPT, to address these concerns by implementing ChatGPT to employ what’s termed **watermarking** and insert certain words more or less often than is customary in even the best of human-authored writing.

To illustrate, let’s assume that most humans use the word **the** an average of 4.8 times per 100 words, whereas ChatGPT might generate prose where **the** appears on average 6.5 times per 100 words. Similarly, humans use the word **of** on average about 3.9 times per 100 words, whereas ChatGPT might leverage **of** about 6.2 times per 100 words. Conversely, ChatGPT might generate the word **by** only 1.6 times per 100 words, whereas humans use the word **by** about 2.7 times per 100 words.

- a. You elect to model word frequency of all words using either a Poisson for paragraphs of 200 or so words—but as a Gaussian for larger documents—say 10000 words or more. Explain why the Gaussian might be the better choice for larger documents than the Poisson, whereas Poisson is more easily defended for smaller documents.

The Poisson distribution provides an accurate estimate for documents with low word counts since we expect mostly low word counts with a long right tail for higher word counts.

As our documents get larger, by the CLT we know that the Gaussian is an accurate, computationally efficient approximate of the word frequencies.

Other answers we accepted were: discussions of variance, discussion of the requirements for binomial approximation, discussion of CLT, discussion of computation efficiency.

- b. A deeper statistical analysis of many human-written documents strongly suggests that $H_{\text{the}} \sim \mathcal{N}(5, 1)$ and $H_{\text{by}} \sim \mathcal{N}(4, 1)$, whereas a separate but equally deep analysis strongly suggests that $C_{\text{the}} \sim \mathcal{N}(3, 1)$ and $C_{\text{by}} \sim \mathcal{N}(2, 1)$. (The parameters are rounded values for simplicity and assumed to be per-100-words.)

Assuming a prior belief that a very long document was written by a human is 0.99, what is your posterior belief that the document was written by a human if the document contains an average of 5 **the**'s every 100 words but only 1 **by** every 100 words. You may assume all Gaussian distributions of interest are independent.

We want to find our posterior belief:

$$f(H|5_{\text{the}}, 1_{\text{by}})$$

By Bayes:

$$= \frac{f(5_{\text{the}}, 1_{\text{by}}|H)P(H)}{f(5_{\text{the}}, 1_{\text{by}})}$$

By LOTP:

$$= \frac{f(5_{\text{the}}, 1_{\text{by}}|H)P(H)}{f(5_{\text{the}}, 1_{\text{by}}|H)P(H) + f(5_{\text{the}}, 1_{\text{by}}|C)P(C)}$$

By our independence, we can expand the joint PDF with a simple product

$$\begin{aligned} &= \frac{0.99 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(0)^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-3)^2}}{0.99 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(0)^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-3)^2} + 0.01 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(2)^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-1)^2}} \\ &= \frac{0.99 \cdot e^{-\frac{1}{2}(-3)^2}}{0.99 \cdot e^{-\frac{1}{2}(-3)^2} + 0.01 \cdot e^{-\frac{1}{2}(2)^2} \cdot e^{-\frac{1}{2}(-1)^2}} \\ &= \frac{0.99 \cdot e^{-\frac{9}{2}}}{0.99 \cdot e^{-\frac{9}{2}} + 0.01 \cdot e^{-\frac{5}{2}}} \end{aligned}$$

which simplifies to about 0.9305.

In reality, the Gaussians here are not independent, since the presence of one word implies the absence of another. Assuming a correlation value ρ that is slightly negative, would you expect the observations stated in part b) to result in a larger posterior probability or a smaller one? Briefly explain why.

We anticipate a larger posterior. Negative correlation implies that an increased usage of "the" is correlated with a decreased usage of "by". This is aligned with what we observed, with "by" being used less than average. Thus, we believe that $f(5_{\text{the}}, 1_{\text{by}}) > f(5_{\text{the}})f(1_{\text{by}})$.